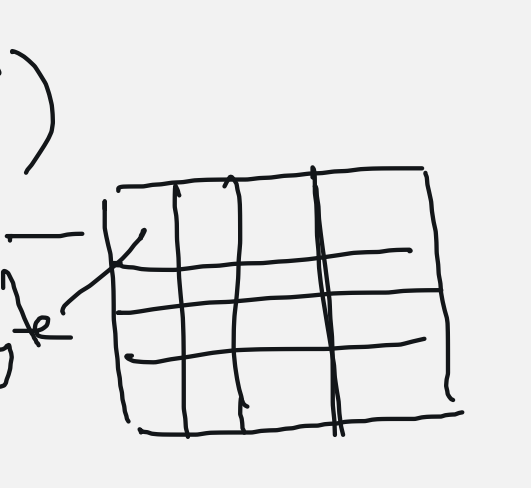
$\mathbb{Z}$, $\mathbb{R}$

integer Real numbers

$(+, -, *, /)$

AES → 128 bits



→ Finite

$\mathbb{Z}_M = \{0, 1, 2, \dots M-1\}$

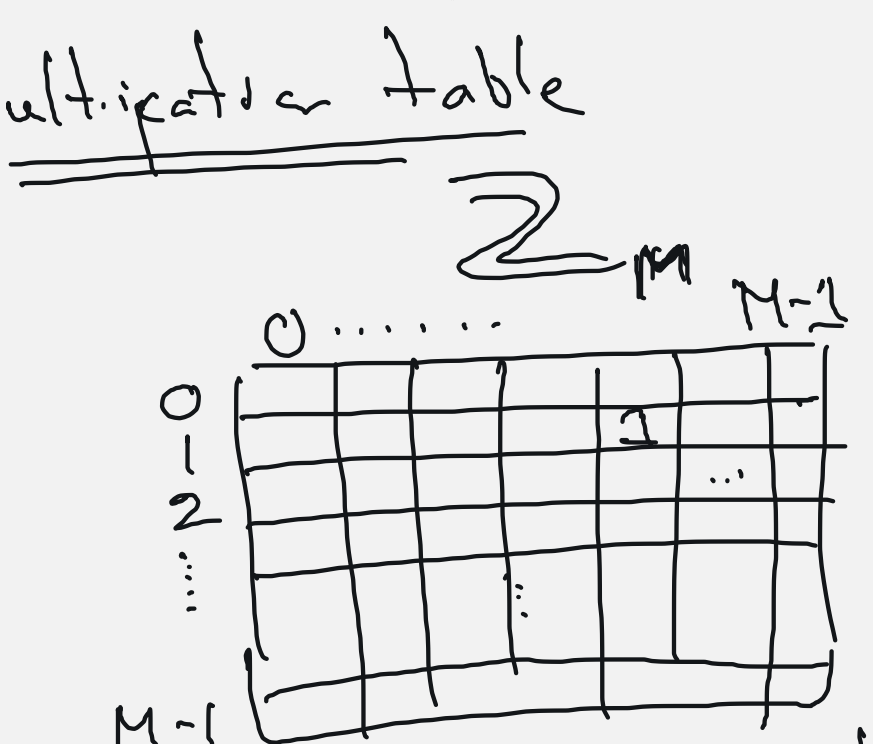
+ modulo M ✓

− modulo M ✓

$*$ ✓

/ → tricky ?

Multiplication table



256 possibilities for byte

$\mathbb{Z}_{256} = \{0, 1, \dots 255\}$

$\mathbb{Z}_4 = \{0, 1, 2, 3\}$ ✓

$[0, 2, 0, 2]$

$a = 2$

$\mathbb{Z}_M = \{0, 1, 2, \dots M-1\}$

M is composite

M =

M = 2

2

$4 = 2 \times 2 = 0$

M = 256, $256 = 128 \times 2$ modulo 256 $= 0$

M = {prime}

M = 3

$\mathbb{Z}_{M|M=3} = \{0, 1, 2\}$ $+, *$ } mod 3

	0	1	2
0	0	0	0
1	0	1	2 ←
2	0	2	1 ←

$2 \times x = 1$

$x = 2$

$1 \times x = 1$

$x = 1$

$0 \times x = 1$

M = 2

	0	1
0	0	0
1	0	1 →

M = 5

$\mathbb{Z}_5 = \{0, 1, 2, 3, 4\}$

	0	1	2	3	4
0	0	0	0	0	0 ✓
1	0	1 ✓	2	3	4 ✓
2	0	2	4	1 ✓	3 ✓
3	0	3	1 ✓	4	2 ✓
4	0	4	3	2	1 ✓

prime → $\mathbb{Z}_M$

→ M to be prime

→ $\mathbb{Z}_M \to (+, -, *, /)$ ✓

$\mathbb{Z}_M = \{0, 1, \dots M-1\}$

Challenge

$\mathbb{Z}_{256} = \{0, 1, 2, \dots 255\}$ → M = 256 is not prime

S_1 S_2 ... S_{256}

00000000 00000001 11111111

option-1 → prime is less than 256 ✓

option-2 → prime is more than 256 —

option 1

$\mathbb{Z}_M = \{0, \dots M-1\}$ ✓ $M < 25\cancel{8}6$

S_1 S_2 ... S_{256}

0 $\mathbb{Z}_{251} = \{0, 1, \dots 250\}$

S_1 S_2 S_3 ... S_{256}

option 2

$\mathbb{Z}_M = \{0, 1, 2, \dots 255, 256\}$ ✓ mod 257

S_1, S_2, ... S_{256}

$S_{14} * S_{19} = S_{24}$

$13 \times 18 = 23$

$S_3 * S_{129} =$

$2 * 128$ − 56 mod 257

$= 256$

option 1 ✗ many to one

option 2 ✗ unused numbers

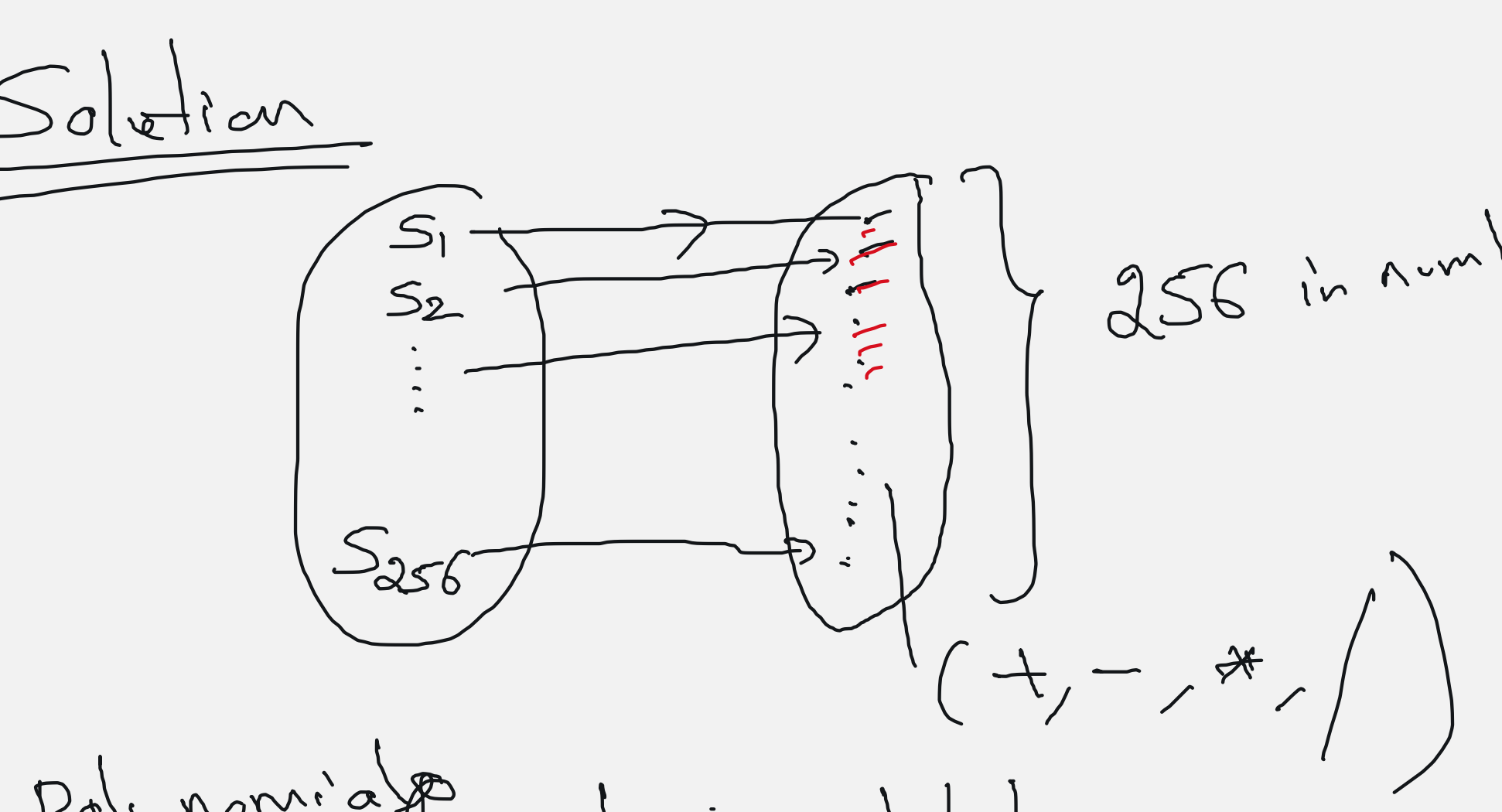
$\{S_1\ S_2 \dots S_{256}\}$

$\{0, 1, 2, 3, \dots 262\}$

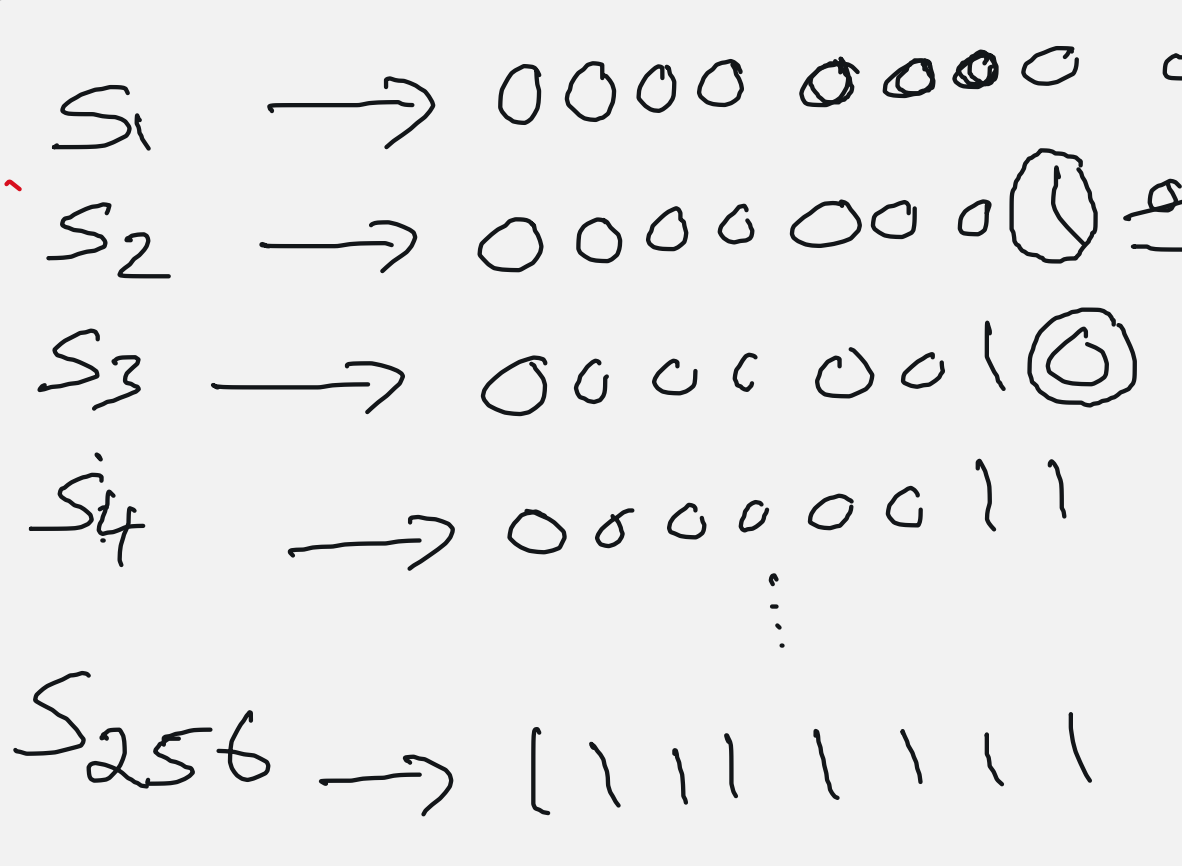
260

$S_3 * S_{131} = ?$ 2

$2 * 130 = 260$



Solution



256 in number

$(+, -, *, /)$

Polynomials

$S_1 \to 0000\,0000$ $0x^7 \dots 0x^2 + 0x + 0 = 0$

$S_2 \to 00000001$ $\cancel{0x^7 \dots 0x^2 + 0x} + 1 = 1$

$S_3 \to 00000010$ $= x$

$S_4 \to 00000011$ $= x + 1$

⋮

$S_{256} \to 11111111$ $\to x^7 + x^6 \dots x + 1$

$01011011 = x^6 + x^4 + x^3 + x + 1$

$\{S_1\ S_2 \dots S_{256}\}$

polynomials of degree at most 7

polynomials of degree at most 1

$\{0, 1, x, 1+x\}$

00 01 10 11

	0	1	x	$1+x$
0	0	0	0	0
1	0	1	x	$1+x$
x	0	x	$1+x$	1
$1+x$	0	$1+x$	1	x

Modulo $x^2 + x + 1$

x^2 modulo $x^2 + x + 1$

$$x^2 + x + 1 \,\big)\, \overline{x^2} \quad 1$$

$$\cancel{x^2} + x + 1$$

$$-x - 1$$

$$= 1 + x$$

$$x^2 + x + 1 \,\big)\, \overline{x^2 + x} \quad 1$$

$$\cancel{x^2} + x + 1$$

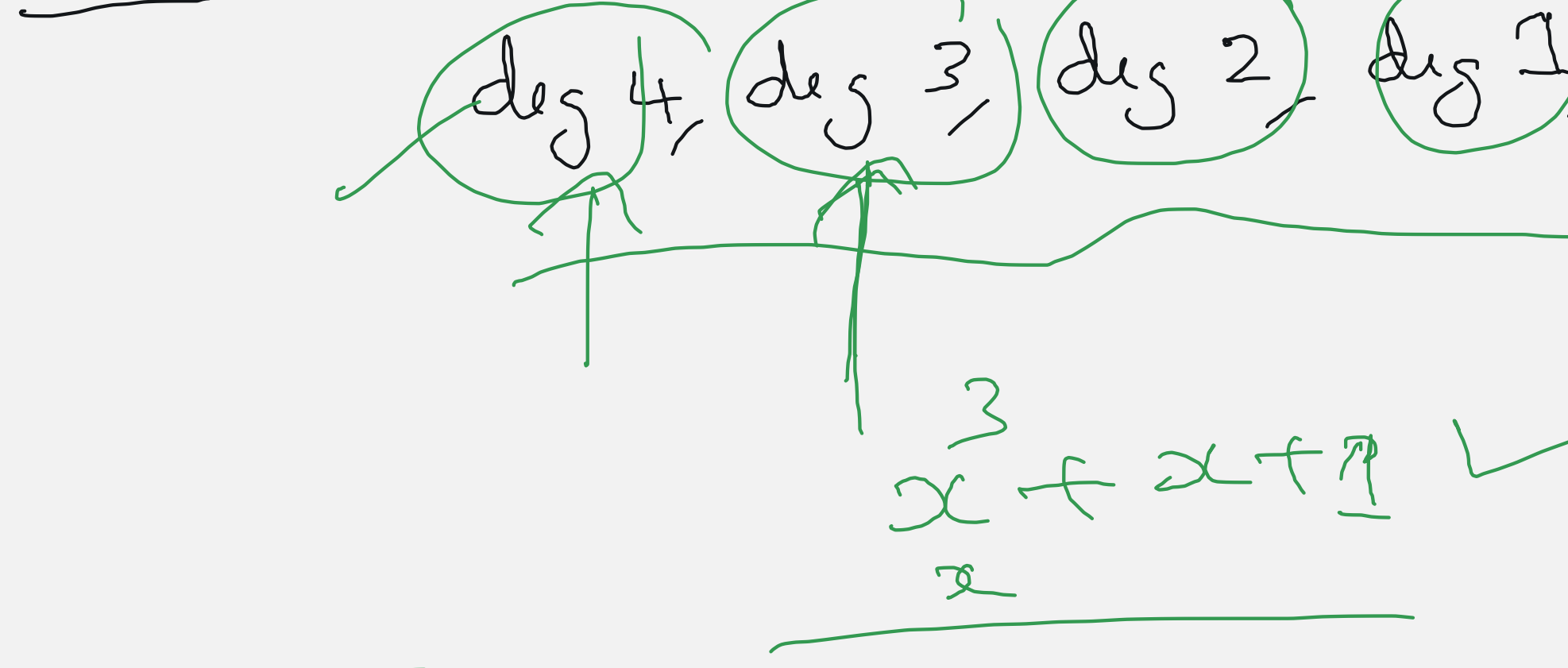
$$-1 = 1$$

prime polynomial

$x^2 + x + 1$ prime poly

3 bits

$\{0, 1, x, 1+x, \dots x^2 + x + 1\}$



Max deg = 4

deg 4, deg 3, deg 2, deg 1, deg 0

$x^3 + x + 1$ ✓

$$x^3 + x + 1 \,\big)\, \overline{x^4} \quad x$$

$$\cancel{x^4} + x^2 + x$$

$$x^2 + x$$

8 bits

$\{0, 1, x, 1+x, \dots x^7 + x^6 + x^5 + x^4 + x^3 + x^2 + x + 1\}$

prime poly → deg(8)